

Online Appendix to

“How much productivity is needed to shorten the workweek? Brazil’s 36-hour proposal and 40-hour alternatives”

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This online appendix collects the data sources, the partial-equilibrium omissions, the calibration details, the sensitivity grids, the sectoral decomposition, the type-level welfare table, the output decomposition, the historical TFP table, the derivations, and the hours-efficiency robustness referenced in the main paper. Section, equation, table, and figure numbers are prefixed with “A.”. Cross-references to the main paper resolve to the labels in the main document.

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A Data sources

All data are from publicly available sources. PNAD Contínua (IBGE/SIDRA), tables 4093, 4097, 6374, 6389, and 6413, provides the informality rate, position in occupation, average hours, published earnings by formal/informal category pair, firm-size distribution, and the CES formal weight $\omega = 0.622$ (Instituto Brasileiro de Geografia e Estatística, 2026). The package freezes the SIDRA pulls in `data_intermediate/pnad_targets.json` and `data_raw/sidra/pnadcm_6389_wage_by_position_last.json`. RAIS/CAGED provides administrative records of formal employment, used for firm-size distribution and formal employment shares (Ministerio do Trabalho e Emprego, 2024). Since the replication package does not call a public RAIS API, the RAIS-derived shares are documented in `data_intermediate/rais_targets.json`. PWT labor-share evidence, adjusted via Gollin (2002), gives $\alpha = 0.35$. The historical TFP anchor uses PWT 11.0 via the FRED series `RTPNABRA632NRUG`, which extends Brazil’s `rtfpna` series to 2023 (Groningen Growth and Development Centre, 2025; Federal Reserve Bank of St. Louis, 2026). The FRED input is frozen at `data_raw/fred/RTPNABRA632NRUG_pwt110.csv`. DIEESE provides contracted-hours categories that the model maps into $\theta = (0.085, 0.269, 0.646)$ for the $\{36, 40, 44\}$ bins (DIEESE, 2024). The mapping assigns the DIEESE ≤ 39 h category to the 36h bin, keeps the 40h category, and assigns the ≥ 41 h category to the 44h bin. An auditing exercise in the replication package verifies that the non-targeted moments in the main text are not algebraic consequences of the calibration.

B What partial equilibrium omits

The static partial-equilibrium framework excludes seven general-equilibrium channels. Capital accumulation responds to the lower marginal product of capital after the effective labor contraction. Wages and prices adjust in general equilibrium, partially restoring formal labor demand. Labor force participation may rise with leisure. Monetary and fiscal authorities may accommodate. Workers may reallocate across sectors. The formalization decision may move in either direction. Work-sharing is shut down by the fixed- N assumption. All seven channels except the last have a negative expected sign on A_{req} .

Table A.1 separates channels already modeled from those bracketed quantitatively and those discussed qualitatively. The table is deliberately balanced: it includes channels that could lower the output-neutrality benchmark, such as capital adjustment, and channels that could raise or redirect it, such as compliance, overtime scheduling, wage bargaining, and organizational disruption.

Table A.1: Modeled, bracketed, and omitted channels around the core partial-equilibrium benchmark.

Channel	Expected sign	Horizon	Object affected	Status
Formal-informal substitution	+ on informality; net model outcome	Short run	A_{req} , informality, incidence	modeled
Capital adjustment	− on A_{req}	3-5 years	A_{req} , output neutrality	bracketed
Below-40 coordination losses	+ on A_{req}	Short run	A_{req} , output	bracketed
Compliance, evasion, PJ/MEI	±	Immediate to short run	Policy bite, informality, incidence	discussed
Overtime premia and scheduling	±	Immediate to short run	Output, labor costs, incidence	discussed
Wage bargaining and prices	±	Short to medium run	Incidence, formal demand, real wages	discussed
Labor supply and work-sharing	usually − on A_{req} ; welfare ±	Medium run	Output, welfare, employment	discussed
Sectoral reallocation and IO links	±	Medium run	Sectoral A_{req} , aggregate output	modeled

Notes: Sign is the expected direction for the effective required productivity gain or for the affected incidence margin, holding the statutory cap fixed. Status takes three values: *modeled* when the channel is inside the quantitative framework, *bracketed* when the paper reports an explicit bound or specification bracket, and *discussed* when the channel is acknowledged qualitatively. The JSON and CSV versions are written to `output/validation/omitted_channels.*`.

The pessimistic TFP stress test lets post-reform TFP fall by δ before computing the productivity gain required to restore output. It is not a forecast of implementation

disruption. It is a robustness exercise for the case in which reorganization, compliance frictions, or learning losses make the transition temporarily less productive.

Table A.2: Pessimistic stress test: reform with post-reform TFP shock.

Post-reform TFP shock δ (%)	A_{req} (%)	ΔY (%)	ΔInf (pp)	ΔCV (%)
0.0 (baseline)	+8.18	−8.02	+1.917	−3.63
0.5	+8.73	−8.51	+1.980	−4.26
1.0	+9.28	−9.00	+2.044	−4.89
1.5	+9.83	−9.50	+2.122	−5.54
2.0	+10.39	−9.99	+2.185	−6.17

δ represents a post-reform contraction of TFP beyond the mechanical effect of the hours cap. The $\delta = 0$ row reproduces the main result of Table 4. Other rows show how outcomes deteriorate if the reform itself depresses TFP by δ .

The capital-adjustment bound in Equation (6) of the main paper uses a textbook user-cost elasticity $\varepsilon_K \in [0.2, 0.4]$ (Caballero, 1999) and the baseline capital share $\alpha = 0.35$. At these elasticities over a 3 to 5 year horizon, the effective A_{req} under the two-sided penalty specification falls from 8.18% to between 6.4% and 7.2%, and under the one-sided fatigue specification to between 4.8% and 5.6%. The remaining six channels are individually small and of mixed sign. The headline numbers should be read as short-run upper bounds on the required TFP gain. The 1 to 2 year model horizon is not the same as the historical PWT window used to anchor A_{req} . The anchor is best read as the cumulative TFP gain required over a 3 to 10 year phase-in horizon.

C Calibration details

C.1 Hours-output elasticity and efficiency shape

The Pencavel evidence documents declining marginal product of hours at long workweeks for a specific industrial setting. Collewet and Sauermann (2017) find analogous long-hour productivity losses in a Dutch call-center panel. Bick et al. (2018) provide cross-country evidence consistent with declining marginal product of hours, though their primary focus is the relationship between hours and income. Neither paper identifies a precise peak location or an efficiency response below a peak. The choice to place the peak at $h^* = 40$ and to compare a one-sided fatigue specification (no penalty for shorter hours) with a two-sided penalty specification (equal curvature either side) is a modeling choice loosely informed by this evidence rather than a direct estimate from it. Section J reports four alternative shapes.

C.2 Informal efficiency and adjustment costs

The informal efficiency $\eta_I = 0.40$ is set at the midpoint of the informal-to-formal wage ratio range ($1/3$ to $1/2$) reported by La Porta and Shleifer (2014), complemented by the Mexican estimates in Busso et al. (2012). Under competitive wage-taking the CES first-order conditions identify η_I through the wage (marginal-product) ratio rather than through value-added per employee, which conflates productivity differences with capital-per-worker gaps between formal and informal firms. The adjustment costs $\gamma_F^S = 0.12$ and $\gamma_F^L = 0.03$ reflect the effective marginal compliance cost of CLT/FGTS obligations for small and large firms. Sensitivity over $\gamma_F^S \in [0.08, 0.16]$ and $\gamma_F^L \in [0.01, 0.05]$ shifts A_{req} by less than ± 0.2 pp. The convex informality cost π_m is calibrated endogenously to ensure non-negative wedges at the informality target.

C.3 Reconciliation with higher substitution elasticities

Refugee-crisis studies report substitution elasticities between worker types of $\sigma \approx 11$ to 30 (Ottaviano and Peri, 2012; Bahar et al., 2025). These magnitudes are not directly comparable to σ_{sub} as defined here. Refugee studies measure substitution between workers of different origin in a market-clearing equilibrium. Here, σ_{sub} captures intra-firm substitution between formal and informal employment regimes. The horizon also differs, since refugee studies identify medium-run responses that include firm entry and organizational reallocation, while the model here holds the firm distribution fixed. The unit differs as well, since refugee studies typically include self-employed workers, while the aggregator concerns salaried employment. DSGE models of informality (Leyva and Urrutia, 2020; McKiernan, 2021) similarly use σ parameters that govern sectoral mobility rather than

intra-firm substitution. In Ulyssea (2018), formal and informal workers are effectively perfect substitutes within the firm, with productivity differences captured entirely by the distribution of firm types. The calibration $\sigma = 1.326$ with $\eta_I = 0.40$ can be read as a parametric decomposition in which the formal-informal wage ratio pins the per-efficiency-unit relative productivity of informal labor, while modest intra-firm substitutability ($\sigma > 1$) absorbs the remaining reallocation margin. Both are disciplined by the observed wage premium.

C.4 Disciplining σ_{sub} : full results

For each σ in a grid over $[0.40, 3.00]$, the full model is recalibrated and the implied formal-informal wage premium is computed from the CES first-order condition,

$$R(\sigma) = \frac{\omega}{1 - \omega} \cdot \left(\frac{L_F}{L_I} \right)^{\rho-1} \cdot z,$$

where $z = \text{eff}_{h_F}/(\eta_I h_I e(h_I))$. The mapping is monotone in σ . At the recalibrated $(\omega, \eta_I) = (0.622, 0.40)$, setting $R \in [1.15, 1.55]$ delivers $\sigma \in [1.116, 1.469]$ and the conservative midbox $A_{\text{req}} \in [7.50\%, 8.58\%]$ (the full (σ, ω, η_I) -box envelope is $[6.93\%, 9.23\%]$). Derenoncourt et al. (2025) estimate a small formal-to-informal reallocation elasticity of about -0.28 from the Brazilian minimum-wage passthrough, consistent with the modest intra-firm σ used in this paper.

The wage-premium bridge compares current published PNAD earnings with the MNR window. These ratios are not used as controlled estimates of the formal premium. They check whether the private-employee and domestic-worker category pairs place current Brazilian formal/informal earnings gaps close to the structural window used to discipline σ_{sub} . The self-employed and employer pairs are shown for context but not used for identification because their earnings mix labor income with capital, scale, and client composition.

Table A.3: Published PNAD earnings ratios and the MNR wage-premium window.

Category pair	Formal (R\$)	Informal (R\$)	Ratio R	Interpretation
Private employees	3,305	2,658	1.24	Most comparable employee pair; inside MNR
Domestic workers	1,931	1,226	1.58	Low-wage employee pair; outside MNR
Public employees	5,035	3,129	1.61	Institutional comparison; outside MNR
Employers	10,119	5,238	1.93	Includes capital/firm scale; outside MNR
Own-account	5,078	2,223	2.28	Includes capital/client mix; outside MNR
MNR structural premium window used in calibration			[1.15, 1.55]	—

Notes: Formal and informal columns report published monthly real earnings from PNAD Continua Mensal/SIDRA Table 6389, variable 5932, dez-jan-fev 2026. The ratios are descriptive category-pair diagnostics, not controlled microdata regressions. They are used only to show how the structural window $R \in [1.15, 1.55]$ from Meghir et al. (2015) relates to current Brazilian published wage gaps. Employer and own-account ratios mix labor income with capital, scale, and client composition and therefore are not used to identify σ_{sub} .

D Sensitivity

The sensitivity analysis varies each parameter within economically disciplined ranges. Three parameters carry most of the variation in A_{req} , namely σ , ω , and η_I . The remaining primitives shift A_{req} by at most ± 0.3 pp. The full (σ, ω) grid leaves the qualitative finding unchanged, with A_{req} above 4% across all combinations in the joint grid.

Table A.4: Univariate sensitivity within economically-disciplined ranges (conservative).

Parameter	Disciplined range	A_{req} range	Hierarchy
σ_{sub}	[1.116, 1.469]	[7.50%, 8.58%]	High (load-bearing)
ω	[0.58, 0.66]	[7.65%, 8.65%]	High
η_I	[0.33, 0.50]	[8.02%, 8.32%]	Moderate
α	[0.30, 0.40]	[7.3%, 8.5%]	Moderate
e_q	[0.40, 0.80]	[7.8%, 8.1%]	Low
h_I	[38, 46]	[7.9%, 8.1%]	Low
ν	[1.0, 3.0]	ΔCV : [-3.4%, -3.3%]	Low (welfare only)

Notes: Each row varies one parameter inside its data-consistent range and holds all others at baseline. “Disciplined range” is the interval drawn from external evidence (Section C). “ A_{req} range” is the implied interval for the required TFP gain under the conservative $e(h)$ specification at the $44 \rightarrow 36$ h cap. “Hierarchy” classifies each parameter by the width of its A_{req} interval.

Table A.5: A_{req} (%) by σ and ω at $\eta_I = 0.40$ (conservative).

$\sigma \setminus \omega$	0.58	0.622	0.66
1.116	7.00	7.50	7.96
1.326	7.65	8.18	8.65
1.469	8.03	8.58	9.06

Notes: Each cell is the required one-off TFP gain (%) at the $44 \rightarrow 36$ h cap under the two-sided penalty $e(h)$ specification, for the listed (σ, ω) pair at $\eta_I = 0.40$ and all other parameters at baseline. The bolded row $\sigma = 1.326$ is the calibrated central value (corresponding to $R \approx 1.40$, the midpoint of the MNR window). The MNR-consistent interval at the recalibrated $(\omega, \eta_I) = (0.622, 0.40)$ is [1.116, 1.469]. Over the three-way (σ, ω, η_I) box with $\eta_I \in \{0.33, 0.40, 0.50\}$, the envelope across the eight corners is [6.93%, 9.23%].

The explicit eight-corner envelope covers both specifications. The corners span $\sigma \in \{1.116, 1.469\}$, $\omega \in \{0.58, 0.66\}$, and $\eta_I \in \{0.33, 0.50\}$. The table is generated from the validation JSON files in the replication package.

Table A.6: A_{req} (%) at the eight corners of the disciplined (σ, ω, η_I) box.

σ	ω	η_I	One-sided fatigue	Two-sided penalty
1.116	0.58	0.33	5.72	7.06
1.116	0.58	0.50	5.62	6.93
1.116	0.66	0.33	6.49	8.01
1.116	0.66	0.50	6.40	7.89
1.469	0.58	0.33	6.68	8.23
1.469	0.58	0.50	6.33	7.80
1.469	0.66	0.33	7.48	9.23
1.469	0.66	0.50	7.17	8.86
Envelope [min, max]			[5.62, 7.48]	[6.93, 9.23]

Notes: Each row reports A_{req} at the 44 $\rightarrow$ 36h cap for one corner of the (σ, ω, η_I) box, with all other primitives at baseline. The table is generated directly from `output/validation/joint_envelope_flat.json` and `output/validation/joint_envelope.json`.

E Sectoral decomposition

This section applies the national framework separately to three broad sectors (agriculture, industry, services) with sector-specific parameters drawn from PNAD Contínua Q4 2024 microdata (209,311 occupied persons, survey-weighted). Common structural parameters (α , σ_{sub} , ω , κ , h^*) are preserved from the national calibration. Each sector is treated as an independent instance of the national model. The exercise does not model cross-sector mobility or input-output linkages.

Agriculture has the highest calibrated formalization wedge. Its informality rate is 71.8%, nearly twice the national average of 44.2% under the broad IBGE definition. Industry (44.4%) and services (41.2%) are closer to the mean. The calibrated wedges range from $\tau_s = 1.48$ in services to $\tau_s = 5.63$ in agriculture. Payroll-side rural obligations include employer social-security contributions, risk-adjusted accident contributions, and third-party contributions such as SENAR, with reporting rules documented in official payroll guidance (Presidencia da Republica, 1991; Receita Federal do Brasil and eSocial, 2021). The comparison is qualitative, since τ_s is calibrated to match sectoral informality rather than imposed from statutory rates.

Table A.7: Sector-specific parameters and calibrated wedges.

Sector	λ_s	VAB	Inf. rate	θ_{36}	θ_{40}	θ_{44}	Hrs (F)	τ_s
Agriculture	0.076	0.077	71.8%	0.101	0.269	0.630	44.1	5.63
Industry	0.205	0.259	44.4%	0.048	0.417	0.535	42.5	2.10
Services	0.720	0.665	41.2%	0.171	0.410	0.419	40.8	1.48
National	1.000	1.000	44.2%	0.143	0.406	0.451	41.2	—

λ_s : employment share, PNAD Q4 2024. VAB: value-added share, IBGE Contas Regionais 2021. θ_b : share of formal workers in hours bin b . Hours: V4039 (habitual). Informality: broad IBGE definition (VD4009 + V4017 CNPJ). Sources: PNAD/SIDRA and RAIS documentation (Instituto Brasileiro de Geografia e Estatística, 2026; Ministerio do Trabalho e Emprego, 2024).

The sectoral exercise applies the national framework separately to agriculture, industry, and services. The aggregate A_{req} imposes a uniform TFP multiplier on all three sectors and solves by bisection for the value that restores total output. Under the full reform, agriculture requires $A_{\text{req}} = 7.60\%$, industry 8.05%, and services 6.91%, with an aggregate of 7.21%. The aggregate is slightly below the national model’s conservative headline of 8.18% because this exercise uses PNAD habitual hours rather than DIEESE contracted hours. Industry’s A_{req} edges out agriculture’s. At $\sigma > 1$ and $\eta_I = 0.40$, the reallocation friction scales with the formal base, which is larger in industry than in agriculture despite agriculture’s higher baseline informality. All three sectors see informality rise in response to the reform, consistent with mildly substitutable formal and informal labor.

Services accounts for 71.0% of model-implied baseline output and contributes 4.8 of the 7.2 pp aggregate output loss. A sector-specific policy exempting agriculture from

Reform	Sector	A_{req} (%)	ΔY (%)	ΔInf (pp)	Affected (% of sector)	Burden ratio ($A_{\text{req}}/\text{Aff.}$)
44 → 40h	Agriculture	1.8	-2.4	+0.6	17.8	10.1
	Industry	1.6	-1.6	+0.4	29.7	5.4
	Services	1.3	-1.3	+0.4	24.7	5.2
	<i>Aggregate</i>	<i>1.4</i>	<i>-1.4</i>	<i>+0.4</i>	<i>25.2</i>	
44 → 36h	Agriculture	7.6	-9.4	+2.9	25.3	30.0
	Industry	8.0	-7.9	+2.7	52.9	15.2
	Services	6.9	-6.7	+2.1	48.7	14.2
	<i>Aggregate</i>	<i>7.2</i>	<i>-7.2</i>	<i>+2.3</i>	<i>47.8</i>	

Affected: formal workers above the new cap, as % of total sector employment. Burden ratio: A_{req} / affected share, descriptive. Aggregate A_{req} : uniform TFP gain solving $\sum_s Y_s(A(1 + A_{\text{req}}), h_1) = \sum_s Y_s(A, h_0)$ by bisection.

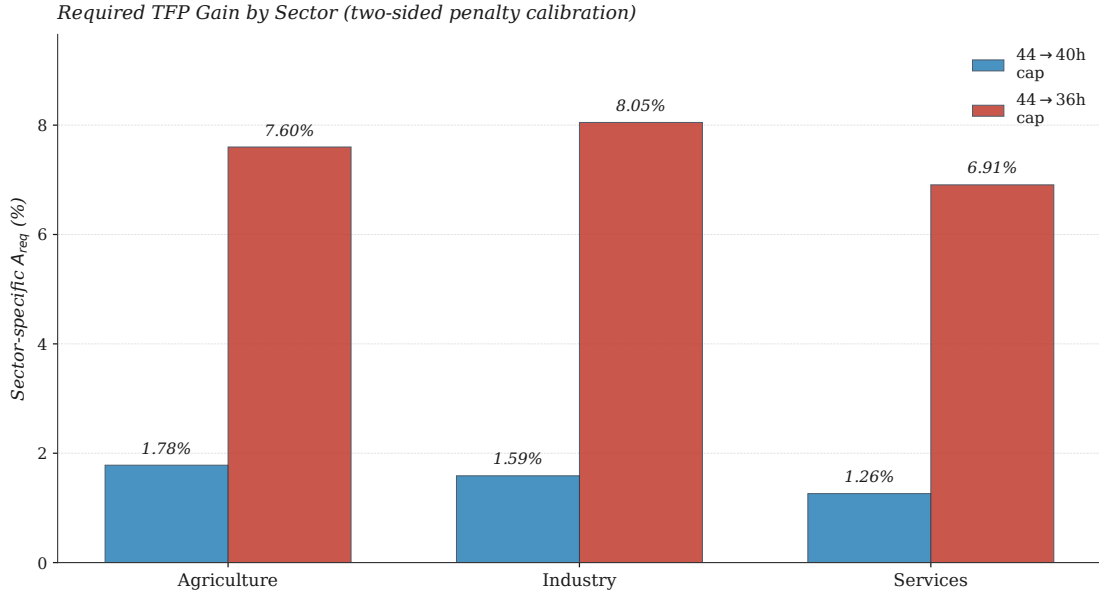


Figure A.1: Required TFP gain A_{req} by sector under the moderate 44 → 40h and full 44 → 36h reforms, conservative calibration.

Notes: Values are computed separately for each sector under the two-sided penalty $e(h)$ specification. Sector-specific parameters (informality, hours distribution, wedge τ_s) come from PNAD Contínua Q4 2024 (Instituto Brasileiro de Geografia e Estatística, 2026). Common structural parameters (α , σ_{sub} , ω , κ , h^*) are kept at their national values.

the cap would reduce aggregate A_{req} by less than 0.2 pp. The decomposition treats each sector as an independent instance of the national model and abstracts from cross-sector reallocation of labor and capital. It also collapses firm-size heterogeneity into a single representative firm per sector.

F Welfare: type-level view

The type-level decomposition uses worker shares and wage-bill shares under the conservative 44 → 36h reform. Workers who remain formal work approximately six fewer hours per week and gain in the first-order incidence calculation. Informal workers' welfare is largely unchanged because $h_I = 44$ is not subject to the cap. The mass of workers de-formalized at the margin faces sharp losses. At the moderate $\eta_I = 0.40$ this mass is small but quantitatively important for incidence. The type-level figures are a first-order approximation. The exact representative-agent CV in Equation (5) of the main paper computes the composite at aggregate consumption and hours, so it is not the weighted sum of the type-level rows.

Table A.9: Type-level welfare incidence at the 36-hour endpoint.

Worker type	First-order ΔCV_k (%)	Worker share (%)	Baseline wage-bill share (%)
Formal stayers	+1.41	60.4	68.5
Informal unaffected	+0.00	37.7	29.3
Deformalized	−28.57	1.9	2.2
Aggregate, representative agent	−3.64	100.0	100.0

Notes: Rows split baseline workers into formal stayers, informal workers already outside the statutory cap, and workers shifted from formal to informal status in the conservative 44 → 36h counterfactual. Formal-stayer incidence uses the GHH parameter from the aggregate calculation plus the CES marginal-product change in formal income. Informal-unaffected workers have zero direct incidence because their hours are not capped. Deformalized workers are assigned the central MNR formal-informal premium $R = 1.40$. The aggregate row is the representative-agent value from the model, not the weighted sum of type-level values.

The separable-preference diagnostic recomputes the welfare schedule using separable log and separable CRRA(2) preferences. The diagnostic is not a distributional welfare analysis, but it checks whether the sign pattern is a knife-edge implication of GHH preferences.

Table A.10: Diagnostic welfare robustness to separable CRRA preferences.

Specification	Hours cap	GHH ΔCV (%)	Sep. log CEV (%)	Sep. CRRA(2) CEV (%)
One-sided fatigue	42h	+0.62	+0.49	+0.48
One-sided fatigue	40h	+0.40	+0.29	+0.27
One-sided fatigue	36h	−1.76	−1.60	−1.81
Two-sided penalty	42h	+0.62	+0.48	+0.48
Two-sided penalty	40h	+0.40	+0.29	+0.27
Two-sided penalty	36h	−3.64	−3.14	−3.43

Notes: GHH reports the model's aggregate ΔCV . The separable columns report baseline consumption-equivalent welfare changes under $u(C, h) = u_\gamma(C) - \chi h^{1+\nu}/(1+\nu)$, with $\nu = 2$ and χ recalibrated from the baseline marginal-rate-of-substitution condition for each γ . The table is diagnostic and does not apply distributional weights.

Figure A.2 shows the welfare response as a function of an exogenous TFP gain at the 36h cap under both specifications. The welfare-neutral thresholds are generated by the replication pipeline and marked in the figure.

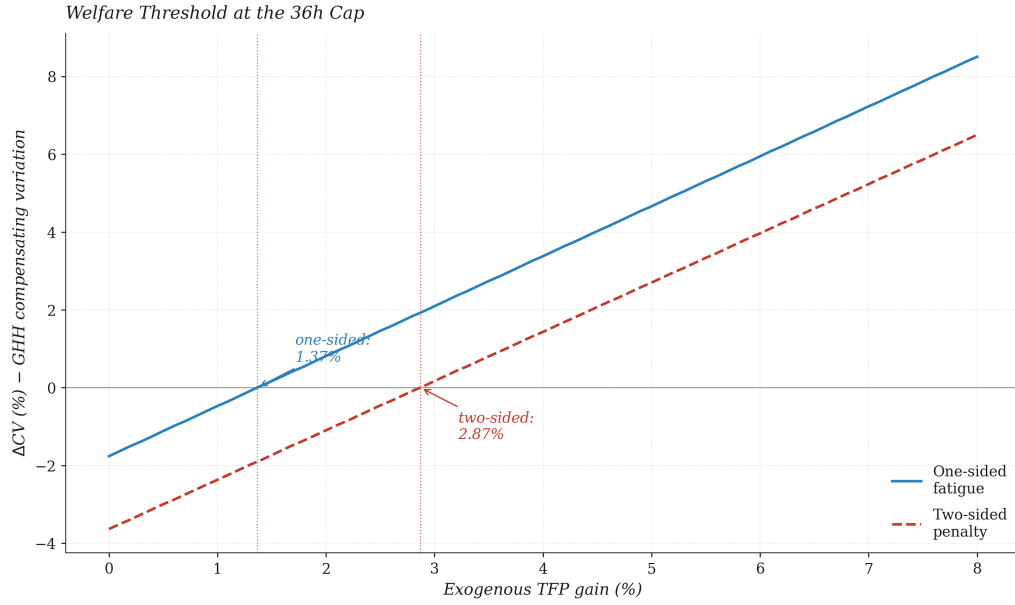


Figure A.2: Welfare ΔCV as a function of TFP gain at the 36h cap.
Notes: Horizontal axis: exogenous one-off TFP gain applied on top of the 44 $\rightarrow$ 36h reform. Vertical axis: aggregate welfare ΔCV from Equation (5) of the main paper. The curves and thresholds are generated by `src/tables_figures/plot_welfare_schedule.py`.

Table A.11: Welfare schedule, two-sided penalty specification.

Hours cap	A_{req}	ΔY	ΔCV	ΔInf
44	0.00%	0.00%	0.00%	0.00 pp
43	0.26%	-0.27%	+0.42%	+0.07 pp
42	0.66%	-0.70%	+0.63%	+0.16 pp
41	1.22%	-1.27%	+0.63%	+0.29 pp
40	1.92%	-2.00%	+0.41%	+0.46 pp
39	3.14%	-3.23%	-0.18%	+0.75 pp
38	4.58%	-4.65%	-1.06%	+1.09 pp
37	6.26%	-6.25%	-2.21%	+1.48 pp
36	8.18%	-8.02%	-3.63%	+1.92 pp
35	10.57%	-10.14%	-5.45%	+2.46 pp

Notes: All rows are model outcomes under the two-sided penalty $e(h)$ specification, measured relative to the 44h baseline. A_{req} is the required TFP gain for the given cap. ΔY is the aggregate output change. ΔCV is aggregate welfare from Equation (5) of the main paper. ΔInf is the change in aggregate informality. The bold row is the 36h endpoint.

G Output decomposition



Figure A.3: Decomposition of ΔY at the 44 → 36h cap under each calibration.

Notes: The bars are generated by `src/tables_figures/plot_main_figures.py`. The mechanical-hours channel is the direct effect of the lower formal-hours cap, holding per-hour efficiency fixed. The efficiency channel is the change in formal hourly efficiency $e(h)$. The residual includes the formal-informal reallocation response and the CES cross-term. The exact values are written to `output/validation/decomposition_results.json`.

H Historical TFP table

Table A.12: Annual TFP gains required to accumulate A_{req} over phase-in horizons.

Scenario	A_{req} (%)	Required annual gain (%/yr)		
		3 yrs	5 yrs	10 yrs
40h alternative, central	1.92	0.64	0.38	0.19
36h endpoint, one-sided fatigue	6.63	2.16	1.29	0.64
36h endpoint, two-sided penalty	8.18	2.66	1.59	0.79

Source: Penn World Table 11.0 via FRED (RTFPNABRA632NRUG), variable **rtfpna**. Required annual gains are $g_T = (1 + A_{\text{req}}/100)^{1/T} - 1$ for $T \in \{3, 5, 10\}$. PWT is used here only as an external scale benchmark: Brazil's annualized PWT growth was -0.83%/yr over 1990-2023 and +0.05%/yr in the best post-1990 rolling decade (2000-2010). These historical series do not validate or reject the policy counterfactual.

I Derivations

I.1 κ from e_q

With $e(h) = \exp\{-\kappa(h - h^*)^2\}$, the hours-output elasticity at h_{ref} is

$$e_q = \left. \frac{d \ln(h \cdot e(h))}{d \ln h} \right|_{h=h_{\text{ref}}} = 1 - 2\kappa h_{\text{ref}}(h_{\text{ref}} - h^*),$$

so

$$\kappa = \frac{1 - e_q}{2h_{\text{ref}}(h_{\text{ref}} - h^*)}. \quad (\text{A.1})$$

With $e_q = 0.60$, $h_{\text{ref}} = 42.244$, and $h^* = 40$, $\kappa = 2.110 \times 10^{-3}$, so $e(36) = e(44) = 0.9668$ and $e(40) = 1$.

I.2 Firm first-order condition

The firm problem has

$$\frac{\partial Y_g}{\partial L_{F,g}} \cdot h_{F,\text{avg}} \cdot e_F \cdot \omega \cdot \left(\frac{L_{F,g}}{L_g} \right)^{\rho-1} = \tau_g + \gamma_{F,g}(N_{F,g} - \bar{N}_{F,g}) + \pi_m N_{I,g}, \quad (\text{A.2})$$

where $\partial Y_g / \partial L_{F,g} = (1 - \alpha)Y_g / L_g \cdot (\partial L_g / \partial L_{F,g})$ and $\partial L_g / \partial L_{F,g} = \omega(L_{F,g} / L_g)^{\rho-1}$.

I.3 A_{req} bisection monotonicity

A_{req} solves Equation (4) of the main paper. The Cobb-Douglas structure $Y_g = AK_g^\alpha L_g^{1-\alpha}$ is linear in A and the optimal $L_g(A)$ is weakly increasing in A , so $\partial \sum_g Y_g / \partial A > 0$ and bisection converges to a unique A_{req} .

J Hours-efficiency robustness

The robustness check compares the one-sided fatigue and two-sided penalty specifications with a no-efficiency benchmark, alternative below-40 curvature, and one-hour shifts in the efficiency peak. Curvature above $h^* = 40$ is fixed at the baseline κ except in the no-efficiency row.

Table A.13: Diagnostics for hours-efficiency assumptions and joint calibration sensitivity.

Specification	$e(36)$	$e(44)$	ΔY (%)	A_{req} (%)	Joint corner range (%)
<i>Main $e(h)$ benchmarks</i>					
No efficiency channel	1.000	1.000	-7.61	7.74	[6.56, 8.73]
One-sided fatigue	1.000	0.967	-6.60	6.63	[5.62, 7.48]
Two-sided penalty	0.967	0.967	-8.02	8.18	[6.93, 9.23]
<i>Compact curvature / h^* sensitivity</i>					
Half curvature below 40h	0.983	0.967	—	7.40	—
Double curvature below 40h	0.935	0.967	—	9.76	—
$h^* = 39$ two-sided	0.987	0.964	-7.05	7.12	—
$h^* = 41$ two-sided	0.909	0.966	-10.51	11.00	—

Notes: The no-efficiency row sets $e(h) = 1$ for all hours by using $\kappa = 0$ and recalibrating the baseline wedges. The one-sided fatigue row uses $e(h) = 1$ for $h \leq 40$ and the baseline fatigue curvature above 40h. The two-sided penalty row applies the same curvature on both sides of 40h. Joint ranges report the eight-corner envelope over $\sigma \in [1.116, 1.469]$, $\omega \in [0.58, 0.66]$, and $\eta_I \in [0.33, 0.50]$.

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